

1. PROBLEM STATEMENT — WHAT MUST BE SOLVED

The problem statement asks for an end-to-end AI-driven mechanism that can identify voice-cloning/synthetic-speech impersonation during live or near-live communication and provide actionable risk information before sensitive actions are taken.

- Near-real-time analysis of incoming voice streams.
- Detection of cloned or AI-generated speech.
- Timely alerts and recommendations before sensitive actions.
- Privacy-preserving processing.
- Scalability across telecom and enterprise environments.
- Multilingual support for Indian languages, accents and dialects.

2. KEY COMPONENTS

For each source-defined key component, the structure below gives the requirement first, followed by the VIVE solution and the approach used to fulfill it.

Key component	What it requires	VIVE proposed solution	How we fulfill it
1. Multi-Layer Voice Authenticity Analysis	Acoustic/spectral analysis; prosody/behavioral analysis; cross-session consistency.	AASIST provides acoustic anti-spoof evidence; ASR + intent/behavior models analyze conversation behavior; ECAPA compares the current speaker with an authorized reference when available.	Audio → VAD/DSP → AASIST + ECAPA → ASR → intent/behavior → combined evidence.
2. Real-Time Risk Scoring Engine	Continuous risk/confidence, configurable thresholds and contextual enrichment.	VIVE combines anti-spoof, speaker, intent, behavior, context, audio quality, uncertainty and temporal evidence into a dynamic risk signal.	Analyze overlapping packets continuously → fuse evidence → update temporal risk → apply policy thresholds.
3. Alerting and User Interaction Layer	UI/SMS/email/in-app alerts, pre-transaction warnings and configurable responses.	VIVE provides live dashboard alerts, mobile/browser notifications and secure webhooks, with recommendations such as secondary verification.	Risk threshold → policy evaluation → authorized notification/workflow.
4. Privacy and Compliance Module	Minimal voice retention, on-device/edge inference and anonymization/feature-only logging.	VIVE is designed around authorized audio, minimal retention, local/edge inference where practical, protected transcripts and feature/metadata logging.	Process only authorized audio → minimize retention → protect sensitive outputs → provide retention/deletion controls.
5. Platform and Integration APIs	REST/gRPC APIs and SDKs for banking, enterprise, contact-center and telecom environments; multilingual support.	VIVE exposes controlled API/SDK boundaries and supports authorized cellular screening plus full-audio analysis through authorized VoIP/in-app/collaboration/media streams.	External system → authenticated session → streaming/evidence API → risk/policy result.

3. EXPECTED OUTCOMES

Expected outcome	Our approach	How the outcome is obtained
1. Reduction in voice-cloning / AI-impersonation fraud	Use multi-evidence detection rather than relying on a single deepfake signal. Combine voice authenticity, speaker consistency, intent, behavior and context before escalating.	Measure event-level detection, false positives/negatives and early-warning performance on defined evaluation data.
2. Improved trust and assurance in voice communication	Show understandable evidence instead of only a final label: risk, confidence, transcript, intent, behavior, contributing signals and risk progression.	Provide a traceable dashboard and packet-level evidence so users can understand why a call became risky.
3. Early detection and proactive containment	Process the call continuously in overlapping windows and update temporal risk while the conversation is active.	Trigger configurable warnings and secondary-verification recommendations as soon as the required evidence threshold is reached.
4. Reusable security layer for telecom and enterprises	Keep communication ingestion, inference, risk and policy behind reusable APIs/SDKs.	Allow the same analysis layer to connect with authorized telecom, banking, enterprise and collaboration workflows.
5. Privacy-preserving and multilingual operation	Keep sensitive audio exposure limited and use Indic-language speech understanding.	Use authorized/local processing where appropriate, minimal retention and Hindi/Tamil-priority ASR with explicit uncertainty for unsupported conditions.

4. ONE-LINE SYSTEM FLOW

Authorized communication → secure ingestion → streaming packets → VAD/DSP → voice authenticity → speaker consistency → ASR → intent → social-engineering behavior → context → uncertainty → risk fusion → temporal risk → policy → dashboard/alerts/APIs.

5. PROJECT POSITIONING

- VIVE is not only a voice-cloning detector.
- Synthetic-voice evidence and scam/impersonation risk are separate signals.
- A real human voice can still be used for social engineering.
- A synthetic-voice signal is not automatically proof of fraud.
- The output is decision support with explainable evidence and authorized actions.